

EDUCATION

New York University

Ph.D., Neural Science

Thesis: Hidden State Inference Modulates Reward Learning but Not Striatal Dopamine Release.

Advisor: Dr. Christine Constantinople

New York City, NY

Aug. 2018 – May 2024

Washington and Lee University

B.S. with Honors, Neuroscience and Mathematics, Magna Cum Laude

Lexington, VA

Aug. 2014 – May 2018

EXPERIENCE

Flatiron Research Fellow

Center for Computational Neuroscience, Flatiron Institute

New York City, NY

Jan. 2025 - Present

- Co-derived theoretical bounds on the error of linear recurrent networks approximating Bayesian filtering in Hidden Markov models.
- Validated the theoretical predictions by training linear recurrent networks on 230 randomly generated and adversarial hidden Markov models using custom JAX and Slurm pipelines on GPU clusters.
- Identified a low-dimensional hidden-state drift associated with failures of length generalization and developed tools for stabilizing model representations, maintaining high performance for sequence up to 100x training length.
- Helped organize a Neuro-AI research subgroup of up to 10 researchers, organizing project discussions, technical presentations, and collaborations across neuroscience and machine learning.

Graduate Research Scientist

Center for Neural Science, New York University

New York City, NY

June 2019 - Dec. 2024

- Led behavioral, computational, and neural studies of how animals infer hidden task states during reward-guided decision-making, resulting in two first-author publications.
- Developed standardized MATLAB and MySQL software for processing multimodal behavioral and neural data, achieving adoption by 11+ lab members and supporting analyses in 6+ peer-reviewed publications.
- Awarded a competitive National Institute of Mental Health F31 predoctoral fellowship supporting research on hidden-state inference in rodents.

PUBLICATIONS

Toporikova, N., Cheng, W., Qian, L., **Mah, A.**, Clarke, T., Jones, T. C., Moore, D., & Ayoub, N. A. (*In Press*). Phase response curve and RNA-sequencing demonstrate spiders' sensitivity to light and pinpoint candidate light-responsive genes. *PLOS ONE*.

Schierack, S. S., Pérez-Rivera, D. T., **Mah, A.**, DeMaegd, M. L., Hocker, D., Ward, R. M., Savin, C., & Constantinople, C. M. (2026). The orbitofrontal cortex updates beliefs for state inference. *Neuron*, 114(3), 507–520. DOI: 10.1016/j.neuron.2025.10.024

Golden, C. E., Martin, A. C., Kaur, D., **Mah, A.**, Levy, D. H., Yamaguchi, T., Lasek, A. W., Lin, D., Aoki, C., & Constantinople, C. M. (2025). Estrogen modulates reward prediction errors and reinforcement learning. *Nature Neuroscience*, 1–13. DOI: 10.1038/s41593-025-02104-z

Mah, A., Golden, C. E., & Constantinople, C. M. (2024). Dopamine transients encode reward prediction errors independent of learning rates. *Cell Reports*, 43(10). DOI: 10.1016/j.celrep.2024.114840

Mah, A., Schierack, S. S., Bossio, V., & Constantinople, C. M. (2023). Distinct value computations support rapid sequential decisions. *Nature Communications*, 14(1), 7573. DOI: 10.1038/s41467-023-43250-x

Mah, A., Ayoub, N., Toporikova, N., Jones, T. C., & Moore, D. (2020). Locomotor activity patterns in three spider species suggest relaxed selection on endogenous circadian period and novel features of chronotype. *Journal of Comparative Physiology A*, 206(4), 499–515. DOI: 10.1007/s00359-020-01412-y

PRESENTATIONS AND CONFERENCES

- Mah, A.,** Williams, A.H.. (2025) Linear Recurrent Networks Approximate Optimal Filtering in Hidden Markov Models. UniReps Workshop, Neural Information Processing Systems. San Diego, California. Poster
- Mah, A.,** Golden, C.E.M., Jang, H., Constantinople, C.M. (2024) Mesolimbic dopamine encodes reward prediction errors independent of learning rates. COSYNE. Lisbon, Portugal. Poster
- Mah, A.,** Golden, C.E.M., Constantinople, C.M. (2023) Neural Instantiation of a Dynamic Learning Rate. Society for Neuroscience. Washington, D.C. Poster.
- Mah, A.,** Constantinople, C.M. (2022) Optimists and realists: heterogeneous priors in rats performing hidden state inference. COSYNE. Lisbon, Portugal. Poster.
- Mah, A.,** Schiereck, S., Bossio, V., Constantinople, C. (2021). Rats infer hidden-states in a temporal wagering task. Society for Neuroscience. Poster.
- Mah, A.,** Li Y., Constantinople, C. (2021). Using neuromodulatory dynamics to arbitrate between distinct behavioral strategies. Society for Neuroscience Global Connectome: A Virtual Event. Poster
- Mah, A.,** Ayoub, N., Jones T.C., Moore, D., Toporikova, N. (2017). Computational model of circadian rhythms predicts adaptive value of exceptionally short period spider clock via flexible entrainment. Society for Neuroscience. Poster.
- Mah, A.,** Ayoub, N., Vienneau-Hathaway, J. (2016) Do RTA-clade spiders possess the same suite of silk genes as orb-web weaving spiders? International Congress of Arachnology. Poster.

GRANTS AND FELLOWSHIPS

Ruth L. Kirschstein National Research Service Award (NRSA) Predoctoral Fellowship, NIMH.	2023 - 2025
2T32MH019524 – Training Systems and Integrative Neuroscience.	2021 - 2023
5T90DA043219 – Training Program in Computational Neuroscience.	2019 - 2021
Henry Mitchell McCracken Fellowship, New York University.	2018 - 2021
The Jarrard Award for Distinguished Neuroscience Research.	2018
Taylor Scholarship for Excellence in Mathematics.	2017